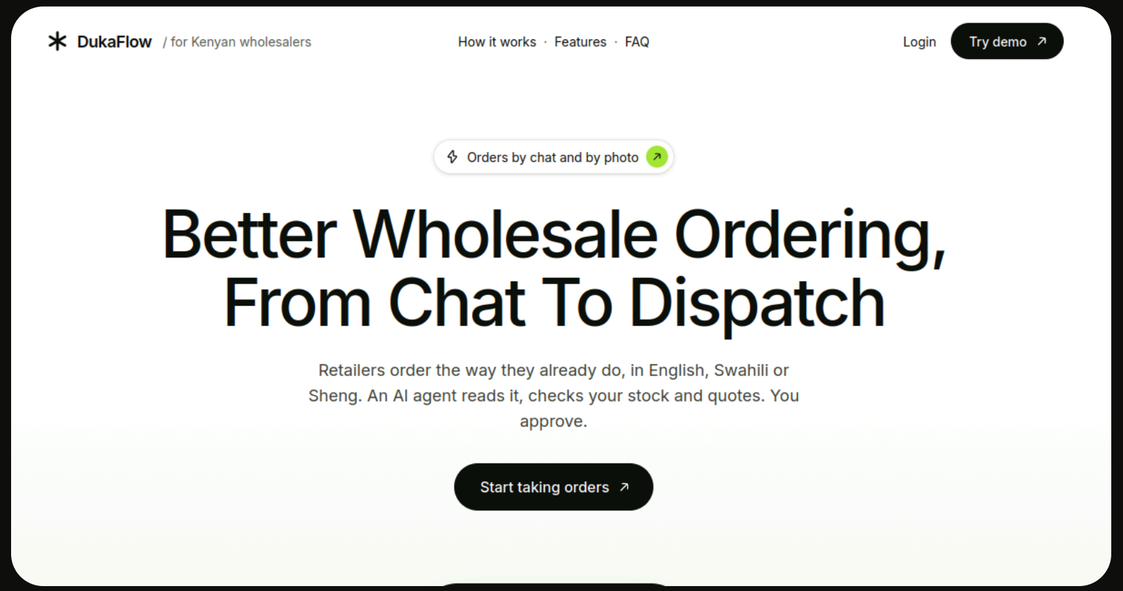
JACHACKS KENYA 2026

DukaFlow AI

An AI order desk for Kenyan wholesalers. Shops order by chat or a photo of a handwritten list; the agent quotes from live stock; the wholesaler approves in one click.



Amon Kipchirchir · github.com/dennisovich254/DukaFlow · Built in Jac

THE PROBLEM

Shops order by text, call and handwritten list. Wholesalers retype it all.

Retyped by hand

“Rina 1L ctn 10, Jogoo bale 5, sukari…” arrives on WhatsApp and is keyed in again.

Stock checked on foot

Shortages show up only when the lorry is being loaded.

Prices from memory

Quotes drift, orders get lost, and mistakes cost margin.

THE SOLUTION

An order desk that never sleeps

1

Chat or photo

English, Swahili or Sheng. No app, no account.

→

2

Agent reads it

Finds products, checks live stock, offers alternatives.

→

3

Priced quote

Catalogue prices, totals by code.

→

4

“Sawa”

The customer confirms; one order is created.

→

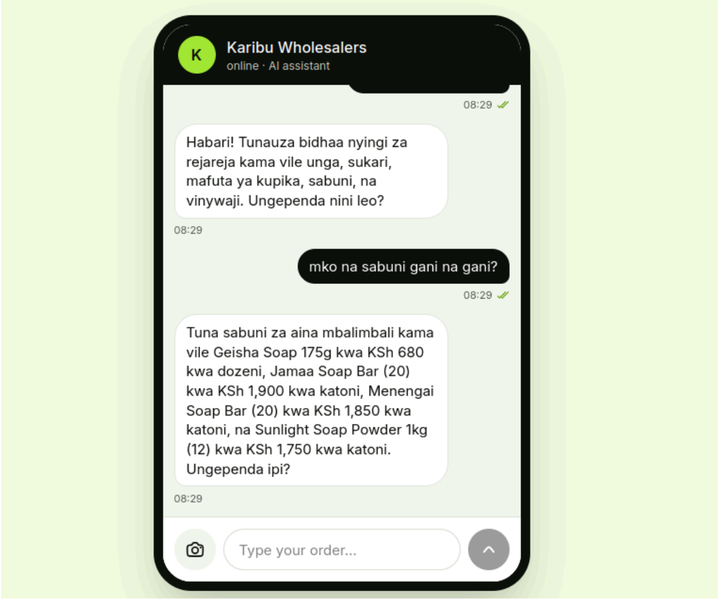
5

Wholesaler approves

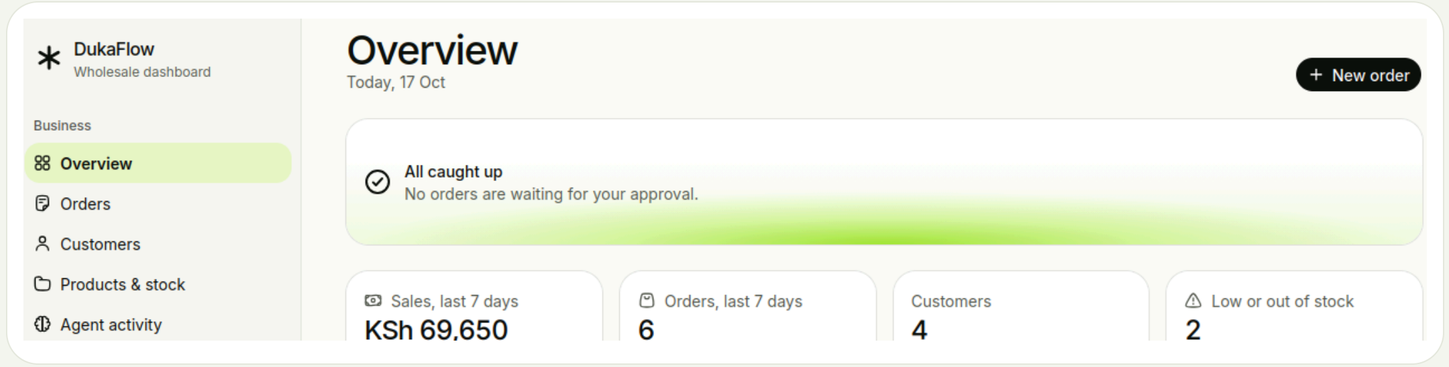
One click, then pack and dispatch.

THE PRODUCT

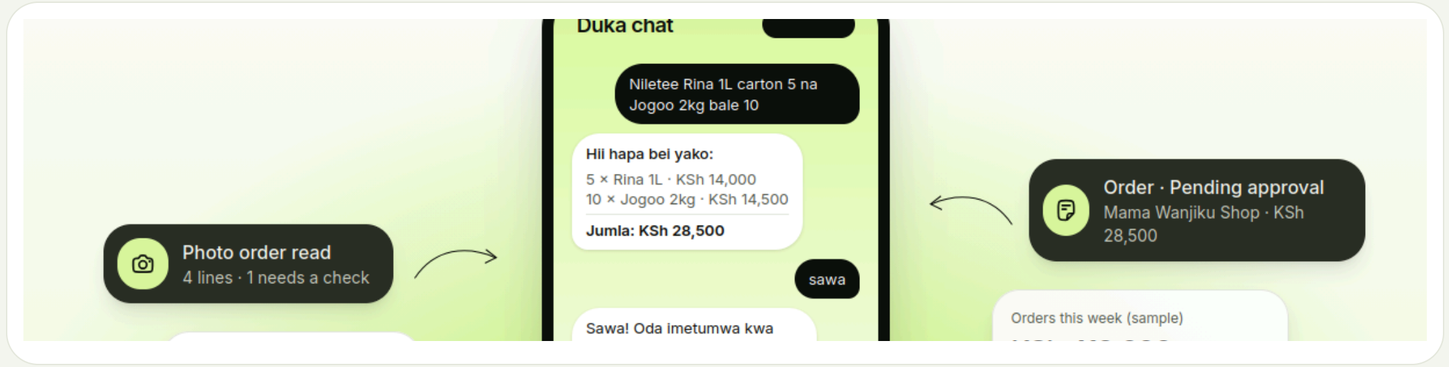
Chat on the phone, control on the dashboard



Shop chat, in the customer's language



Overview: sales, orders and stock at a glance



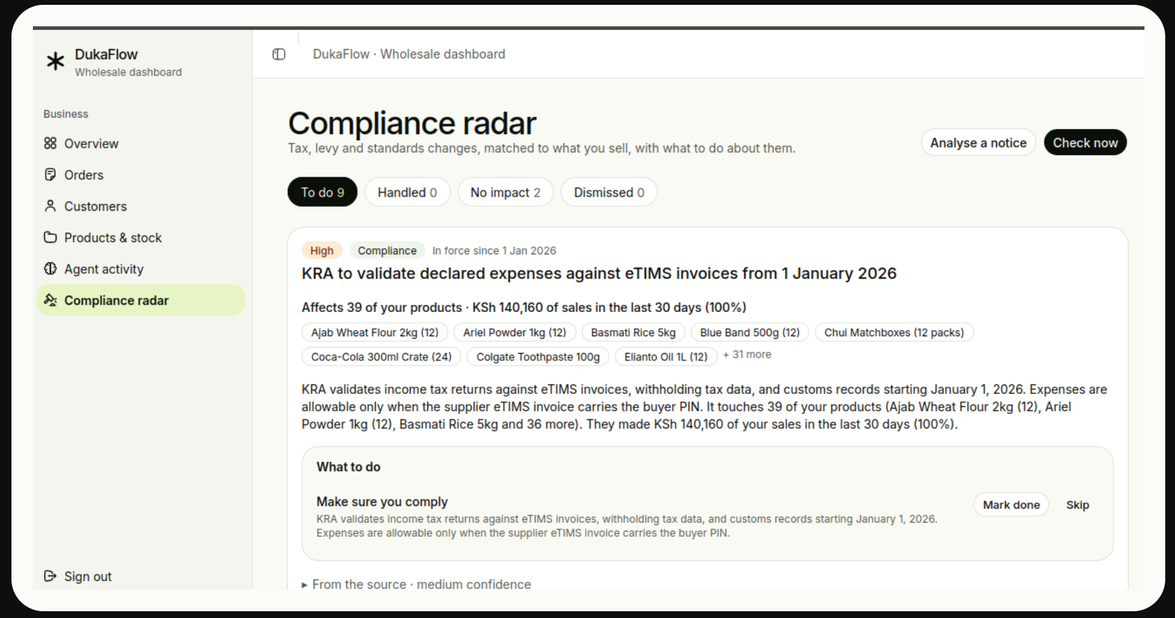
From quote to approval

COMPLIANCE RADAR

New laws, in plain language, for your shelves

Kenya passes bills, levies and standards that few people have time to read, written in jargon like “exempt supplies” and “excisable value”.

The radar reads them, checks every fact against the source, shows which of **your** products are hit and how much of your sales is at stake, then suggests what to do: reprice, stock up, or message affected customers.



AGENTIC DEPTH

The model proposes. Code decides.

Tools

Search catalogue, check stock, suggest alternatives, read past orders.

Memory

“Nipe kama kawaida” quotes the customer’s usual order.

Alternatives

Short on stock? It offers a substitute the customer must confirm.

Plan-act loop

Up to five tool steps, then a typed decision that code validates.

Two agents

The order desk reacts; the Compliance Radar acts proactively.

Human approval

No order, price or message goes out without the wholesaler.

BUILT WITH JAC AND JASECI

The business is a graph; the agents walk it

Jac feature	How DukaFlow uses it
Nodes and edges	Business, products, stock, orders, chats, notices: one persistent graph
Walkers	CustomerAgent walks chat → business; ComplianceScout walks notices → products
by llm with tools, sem	Typed, tool-using decisions; notice reading and fact checking
Image input	Reading photographed handwritten lists
MockLLM, MockToolCall	Agent tests that never call a model
Full-stack client	Every page and component written in Jac
def:pub / def:protect	Four public chat endpoints; everything else needs login

TRUST AND SAFETY

Guardrails a business can rely on

No invented prices

A reply that names a price no tool returned is thrown away.

Facts checked twice

Rates and dates looked up in the source by code; claims judged by a separate model pass.

Invite links, not accounts

Hashed, revocable, 30-day tokens; each reaches only its own chat.

Tested against abuse

Prompt-injection, rate-limit and access tests; a fallback when the model fails.

RESULTS

Built and measured during the hackathon

97.8%

of source code is Jac

130+

automated tests, run by CI on every pull request

11/11

real-model text cases passed in English, Swahili and Sheng

The 11 cases are small and author-written; details in docs/evaluation.md.

WHAT'S NEXT

From one wholesaler to every duka

WhatsApp

The agent already takes channel-independent messages.

M-Pesa

Payment links on approved orders.

Live radar

Scheduled monitoring of official sources, alerts pushed to you.

Code: github.com/dennisovich254/DukaFlow
Video: youtube.com/watch?v=JZkEvwCbrXI
Live: preview-jac-sbx-2bf6dc63c2d54b1b944434e0e53310c3.jachammer.app